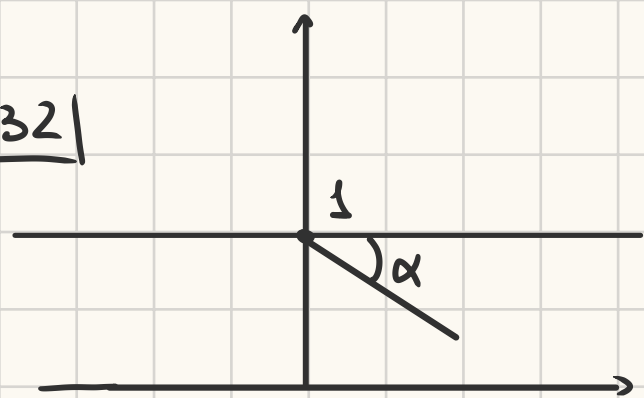


132|



$$\xi = \operatorname{tg} \alpha$$

равномерн распрег: $\frac{\alpha}{\pi}$

$$p(\xi \leq x) = p(\operatorname{tg} \alpha \leq x) = p(\alpha \leq \arctg x)$$

$$F_{\xi}(x) = \frac{\arctg(x) + \pi/2}{\pi} = \frac{\arctg(x)}{\pi} + \frac{1}{2}$$

$$f(x) = (F_{\xi}(x))' = \frac{1}{\pi(x^2+1)}$$

$$E_{\xi} = \int_{\mathbb{R}} f(x) x dx = \frac{1}{\pi} \int_{-\infty}^{+\infty} \frac{x}{x^2+1} dx$$

$$\int_{-\infty}^{+\infty} \left| \frac{x}{x^2+1} \right| dx = 2 \int_0^{+\infty} \frac{x}{x^2+1} dx = \left[\begin{array}{l} u = 1+x^2 \\ du = 2x dx \end{array} \right] = \frac{1}{2} \ln(1+x^2)$$

На бесконечности $\ln(1+n^2) \rightarrow \infty$ — интеграл расходится

$$134| \xi \in E_{0.5} \quad \alpha = \frac{1}{2}$$

$$f(x) = \begin{cases} 0, & x < 0 \\ \frac{1}{2} e^{-\frac{1}{2}x}, & x \geq 0 \end{cases}$$

$$F(x) = \begin{cases} 0, & x < 0 \\ 1 - e^{-\frac{1}{2}x}, & x \geq 0 \end{cases}$$

$$E_{\xi} = \frac{1}{\alpha} = 2 \quad D_{\xi} = \frac{1}{\alpha^2} = 4 \quad p(1 < \xi < 3) = e^{-\frac{1}{2} \cdot 1} - e^{-\frac{1}{2} \cdot 3} \approx 0,3834$$

$$135| \xi \in E_{\alpha}; \quad E_{\xi} = 5 \quad E_{\xi} = \frac{1}{\alpha} \Rightarrow \alpha = \frac{1}{5}$$

$$p(\xi > 10) = 1 - p(\xi \leq 10) = 1 - p(0 \leq \xi \leq 10) = 1 - (e^{-\frac{1}{5} \cdot 0} - e^{-\frac{1}{5} \cdot 10}) = 1 - 1 + e^{-2} \approx 0,14$$

$$F(x) = \begin{cases} 0, & x < 0 \\ 1 - e^{-\frac{x}{5}}, & x \geq 0 \end{cases}$$

$$136| \xi \in N(1,4) \quad a=1 \quad \sigma=4$$

$$p(\alpha < \xi < \beta) = \Phi\left(\frac{\beta-a}{\sigma}\right) - \Phi\left(\frac{\alpha-a}{\sigma}\right) = \Phi\left(\frac{5-1}{2}\right) - \Phi\left(\frac{-1-1}{2}\right) =$$

$$= \Phi(2) - \Phi(-1) = \Phi(2) + \Phi(1) \approx 0,8185$$

137 $\xi \in N(2; 3)$
 $a=2 \quad \sigma^2=3$

$$P(|\xi - 2| < 3) = 2F_0\left(\frac{3}{\sqrt{3}}\right) - 1 = 2\left(0,5 + \Phi\left(\frac{3}{\sqrt{3}}\right)\right) - 1 = 2\Phi\left(\frac{3}{\sqrt{3}}\right) \approx 0,9164$$

$$P(\xi > 0) = P(0 < \xi < +\infty) = \Phi(\infty) - \Phi\left(\frac{0-2}{\sqrt{3}}\right) = 0,5 + \Phi\left(\frac{2}{\sqrt{3}}\right) \approx 0,8749$$

139 $\xi \in N(166; 36)$ $a=166 \quad \sigma^2=36$

$$P(\xi > 154) = P(154 < \xi < +\infty) = \Phi(\infty) - \Phi\left(\frac{154-166}{6}\right) = 0,5 + \Phi(2) \approx 0,9772$$

140 $\xi \in N(a, \xi^2)$

$$P(|\xi| < 42) = 0,92 \Rightarrow P(-42 < \xi < 42) = 0,92 = 2\Phi\left(\frac{42}{\sigma}\right)$$

$$\Phi\left(\frac{42}{\sigma}\right) = 0,46 \quad \frac{42}{\sigma} = 1,75 \Rightarrow \sigma = \frac{42}{1,75} = 24 \text{ mm}$$

142 20×30 1) $N(20, 1^2)$ 2) $N(30, 2^2)$

1) $18-21 \text{ mm}$ $D\xi_1 = 1 \text{ mm}$

2) $27-34 \text{ mm}$ $D\xi_2 = 2 \text{ mm}$

$$P_1 = \Phi\left(\frac{21-20}{1}\right) - \Phi\left(\frac{18-20}{1}\right) = \Phi(1) + \Phi(2) \approx 0,8185$$

$$p = P_1(18 < \xi < 21) \cdot P_2(27 < \xi < 34)$$

$$P_2 = \Phi\left(\frac{34-30}{2}\right) - \Phi\left(\frac{27-30}{2}\right) = \Phi(2) + \Phi\left(\frac{3}{2}\right) \approx 0,9104$$

$$p = 0,8185 \cdot 0,9104 = 0,7452 \quad q = 0,2548$$

$$p(1,5) = 5 \cdot 0,2548 \cdot (0,7452)^4 \approx 0,3931$$

Омбер: 0,3931

143 a) $\xi \in U(a:b)$

$$E\xi = \frac{a+b}{2} \quad D\xi = \frac{(b-a)^2}{12} \quad \sigma = \frac{b-a}{2\sqrt{3}}$$

$$P\left(\left|\xi - \frac{a+b}{2}\right| < \frac{3(b-a)}{2\sqrt{3}}\right)$$

$$P\left(\frac{-3(b-a)}{2\sqrt{3}} < \xi - \frac{a+b}{2} < \frac{3(b-a)}{2\sqrt{3}}\right) = 1$$



8) $\xi \in E_\alpha$ $E\xi = \frac{1}{\alpha}$ $D\xi = \frac{1}{\alpha^2}$ $\sigma = \frac{1}{\alpha}$

$$P(|\xi - E\xi| \leq 3\sigma) = P\left(\left|\xi - \frac{1}{\alpha}\right| \leq \frac{3}{\alpha}\right) = P\left(-\frac{3}{\alpha} < \xi - \frac{1}{\alpha} < \frac{3}{\alpha}\right) =$$

$$= P\left(-\frac{2}{\alpha} < \xi < \frac{4}{\alpha}\right) = P\left(0 < \xi < \frac{4}{\alpha}\right) = 1 - e^{-\frac{4}{\alpha} \cdot \alpha} = 1 - e^{-4}$$

145 $\xi_i \in U(0:1)$ $\eta = \max(\xi_1, \dots, \xi_n)$ $x \in [a:b]$

$$P(\eta \leq x) = P(\xi_1 \leq x, \xi_2 \leq x, \dots, \xi_n \leq x) = x^n \quad F(x) = \begin{cases} 0, & x < 0 \\ x^n, & x \in [0:1] \\ 1, & x \geq 1 \end{cases}$$

$$P(\xi_i \leq x) = \frac{x-0}{2-1} = x \quad f(x) = (F(x))' = n \cdot x^{n-1}$$

$$E\xi = \int_0^1 f(x) \cdot x \, dx = \int_0^1 x^n \cdot n \, dx = n \frac{x^{n+1}}{n+1} \Big|_0^1 = \frac{n}{n+1} \quad \text{Oder: } \frac{n}{n+1}$$